Analytical Cume
January 25, 2019

1. (50 points) Consider the following reaction, in which the starting concentrations are:
$[Cr^{3+}] = [BrO_3^-] = 0.010$ and $[H^+] = 1.0$ that has been allowed to come to equilibrium:

$$BrO_3^- + 2Cr^{3+} + 4\,H_2O \leftrightarrow Br^- + Cr_2O_7^{2-} + 8H^+ \qquad K = 10^{11}$$

You may assume the reaction takes place in a one-liter volume with water.

Find the equilibrium concentrations of all the species.

2. (25 points) The more dilute the acid, the higher the fraction of dissociation, α.
For both parts a &b, you may ignore the contributions of H^+ and OH^- from water
In the solution you are allowed to make no other assumptions.

a. Find the fraction of HA in the form of A^-, in a 0.5 M solution of and acid HA, with a Ka of 1.8×10^{-5}.

$$\alpha_{[A^-]} = \frac{[A^-]}{0.5}$$

b. Find the fraction of HA in the form of A^-, in a 5 μM solution of and acid HA, with a Ka of 1.8×10^{-5}.

$$\alpha_{[A^-]} = \frac{[A^-]}{5X10^{-6}}$$